FleetFlow

Secure Vehicle Rental & Fleet Management System

A complete full-stack web platform for managing vehicles, bookings, payments, maintenance, and fleet operations

React + TypeScript · Node.js + Express · PostgreSQL + Prisma · Tailwind CSS · JWT Authentication · Role-Based Access

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NAVIGATION

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SECTION 1

System Introduction & Overview

A full-stack, production-ready vehicle rental and fleet management platform built with modern web technologies.

FleetFlow is a secure, full-stack web application designed to manage every aspect of a vehicle rental business. It covers the complete operational lifecycle — from browsing and booking a vehicle online, through payment, pickup, active rental, return inspection, and final invoicing — all within a single unified platform.

The system is built for real-world use, with a focus on security, data integrity, and role-appropriate access. It supports four distinct user roles — Administrator, Fleet Manager, Rental Agent, and Customer — each with a tailored interface and strictly enforced permissions on the backend.

What FleetFlow Solves

Traditional vehicle rental businesses often rely on spreadsheets, paper contracts, and disconnected tools. FleetFlow replaces all of that with:

- A **customer-facing booking portal** where clients browse vehicles, check availability, submit bookings, and pay online
- A **staff management console** where agents approve bookings, record pickups and returns, and manage vehicle inspections
- A **fleet operations dashboard** where managers track vehicle utilisation, schedule maintenance, manage drivers, and export reports
- A **secure audit trail** that records every action performed in the system for compliance and accountability

Key Capabilities

Fleet Management
Add, edit, and track every vehicle — availability, compliance dates, mileage, and service history.

Booking System
Real-time availability checking with concurrency-safe double-booking prevention using serializable database transactions.

Payment Flow
Structured simulated payment with security deposit, collateral tracking, and full transaction history.

Maintenance
Schedule and track all service records, assign technicians, and automatically flag vehicles as unavailable during maintenance.

Driver Management
Maintain driver profiles, track licence expiry, manage availability, and assign drivers to bookings.

Reports & Exports
Generate booking and payment reports. Export CSV files with formula injection protection for safe analysis.

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Security & Audit
Argon2id password hashing, JWT tokens, rate limiting, and a complete audit log of all system events.

Contracts
Generate and manage rental contracts tied to confirmed bookings, with full term documentation.

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SECTION 2

Technology Stack

FleetFlow is built entirely with modern, production-grade open-source technologies.

Layer	Technology	Purpose
Frontend Framework	React 18 + TypeScript	Component-based UI with strong typing
Frontend Build	Vite 6	Fast development server and production bundler
Styling	Tailwind CSS 3	Utility-first responsive design system
Forms & Validation	React Hook Form + Zod	Client-side form handling and schema validation
HTTP Client	Axios	API calls with automatic JWT refresh interceptor
Charts	Recharts	Revenue and booking analytics charts
Routing	React Router v7	Client-side navigation with protected routes
Backend Runtime	Node.js + Express.js	REST API server
Backend Language	TypeScript	Type-safe server code
Database	PostgreSQL 14	Relational database with ACID transactions
ORM	Prisma 5	Type-safe database access and migrations
Password Hashing	Argon2id	Industry-leading secure password storage
Authentication	JWT (Access + Refresh)	Stateless token-based auth with revocation
Security Headers	Helmet.js	HTTP security headers
Rate Limiting	express-rate-limit	Brute force and abuse protection
Logging	Winston	Structured server-side logging
Email	Nodemailer + Ethereal	Password reset emails (preview in development)
Monetary Math	Decimal.js	Precise decimal arithmetic for financial calculations
Testing	Vitest	Unit tests for pricing engine and auth utilities
Containerisation	Docker + Docker Compose	Reproducible deployment environment
CI/CD	GitHub Actions	Automated lint, type check, test, and build pipeline

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SECTION 3

Landing Page — Public Interface

The first thing any visitor sees when they open FleetFlow. Fully public, no login required.

When a user navigates to the FleetFlow URL, they are greeted by a professional marketing landing page. This page is the public face of the platform. It communicates the system's value, showcases the fleet, and provides clear entry points for new and returning users. Visitors who are already logged in are automatically redirected to their dashboard.

Hero Section: Landing Page

The full-screen hero section shows a dramatic car photograph with a dark overlay. The headline reads "Drive your fleet forward." in large bold text with the phrase highlighted in blue. Two call-to-action buttons are displayed prominently: **Get started free** (blue, primary) which takes new users to the registration page, and **Sign in** (dark, secondary) for returning users. Trust badges showing "500+ Vehicles", "Real-time Tracking", and "Secure Platform" build credibility. The navigation bar at the top contains links to Features, Fleet, and About sections, plus Sign in and Get Started buttons.

Features Section: "Everything you need"

This section lists the six core capabilities of FleetFlow displayed as animated cards. Each card has a coloured icon and a short description: Smart Fleet Management, Secure & Compliant, Instant Bookings, Powerful Analytics, Multi-Role Access, and Full Rental Lifecycle. Cards animate upward as the user scrolls down the page.

Fleet Showcase: "Every vehicle, every need"

Four vehicle cards show the BMW 5 Series (Luxury), Ford Explorer (SUV), Toyota Corolla (Economy), and Toyota RAV4 (Hybrid SUV). Each card shows the category badge, a full-width photo, the vehicle name, the daily rental price in a blue badge, and a "Book this vehicle" button. When hovered, the card lifts upward and the photo zooms in slightly. Clicking any card takes the visitor to the registration page.

How It Works: 3-Step Process

A three-step visual guide explains the rental process. **Step 01 – Create account:** Sign up in seconds, no credit card required. **Step 02 – Browse & book:** Choose from the fleet, set your dates and submit a booking request. **Step 03 – Pick up & drive:** Get approved, collect the keys, enjoy the rental and return. Below this section is a call-to-action banner inviting visitors to join FleetFlow.

CTA Banner & Footer

A deep blue banner with a car photograph in the background contains the headline "Ready to get started?" with two buttons: **Create free account** (white) and **Sign in** (outlined). This is the final conversion point on the landing page. The footer shows the FleetFlow logo, copyright information, and quick links to Sign in and Register.

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SECTION 4

Authentication — Login & Registration

Secure access control with Argon2id hashing, JWT tokens, and rate limiting.

FleetFlow uses a modern JWT-based authentication system. When a user logs in, they receive a short-lived access token (15 minutes) and a long-lived refresh token (7 days). The access token is automatically refreshed in the background so users are never interrupted mid-session. Passwords are hashed using Argon2id — the strongest password hashing algorithm available — with 64 MB memory cost and 3 iterations.

> **Security note:** Public registration always creates a Customer account. It is not possible to self-register as an Administrator or Fleet Manager. Staff accounts are created by administrators only.

Login Page: Sign In

The login page uses a split-screen layout. The left panel shows a full-screen photograph of a sports car with the tagline "Drive your fleet forward." and trust badges. The right panel is clean white with a "Sign in" heading, email address input, password input with a show/hide toggle, a "Forgot password?" link, and a blue "Sign in" button. A "← Back to home" link returns the user to the landing page. Authentication attempts are rate-limited to prevent brute-force attacks.

Registration Page: Create Account

The registration page follows the same split-screen design, with a different car photograph on the left and the text "Join thousands of happy renters." The form collects first name, last name, email address, phone number (optional), password, and confirm password. The password must be at least 8 characters with at least one uppercase letter, one lowercase letter, one number, and one special character. All new accounts are created with the Customer role.

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SECTION 5

User Roles & Permissions

FleetFlow has four distinct roles, each with a tailored interface and strictly enforced backend permissions.

Role-based access control (RBAC) is enforced on every API endpoint on the server. The frontend interface simply reflects what each user is allowed to do — hiding a button does not grant access. If a user attempts to call a restricted API endpoint directly, the server rejects the request with a **403 Forbidden** response.

Administrator • Highest Access

The Administrator has full access to every module in the system. They can manage all users, vehicles, bookings, payments, contracts, maintenance, drivers, and reports. They are the only role with access to the Audit Log, which records every action performed in the system. Administrators can change user roles, suspend accounts, and create staff members.

Sidebar modules: Dashboard, Vehicles, Bookings, Contracts, Customers, Payments, Maintenance, Drivers, Reports, Notifications, Audit Log

Fleet Manager • Operations

The Fleet Manager oversees the vehicle fleet and operational side of the business. They can add and edit vehicles, schedule maintenance, manage drivers, view revenue charts, and export reports. They can also approve bookings and manage contracts, but they cannot access the Audit Log or change user roles.

Sidebar modules: Dashboard, Vehicles, Bookings, Contracts, Customers, Payments, Maintenance, Drivers, Reports, Notifications

Rental Agent • Front Desk

The Rental Agent handles the day-to-day customer interactions. They can approve or reject bookings, record vehicle pickups and returns, verify customers, create contracts, record payments manually, and manage maintenance requests. They cannot modify vehicle pricing, access financial reports, or view the audit log.

Sidebar modules: Dashboard, Vehicles, Bookings, Contracts, Customers, Payments, Maintenance, Drivers, Notifications

Customer • Self-Service

The Customer can browse available vehicles, submit booking requests, make payments for approved bookings, and view their own rental history. They cannot access any other customer's data, modify prices, or access any staff module. All data access is strictly limited to records they own.

Sidebar modules: Dashboard, Vehicles, Bookings, Payments, Notifications

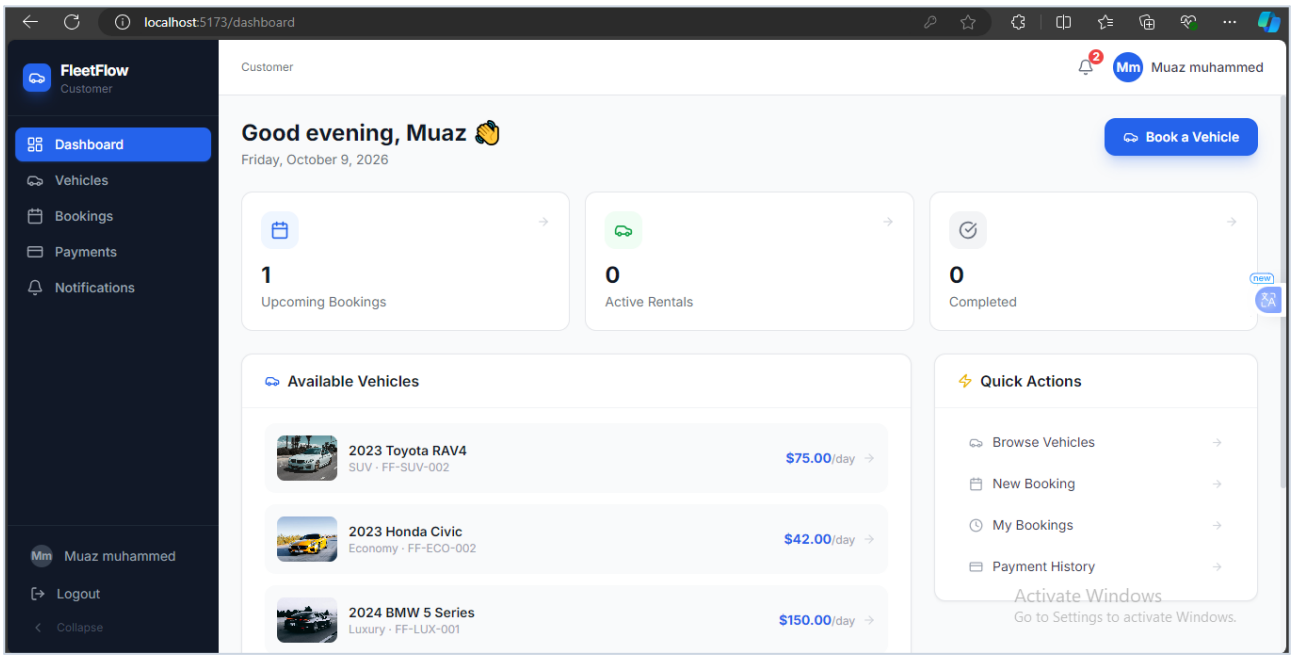
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SECTION 6

Customer Role — Screens & Features

The self-service experience for vehicle renters.

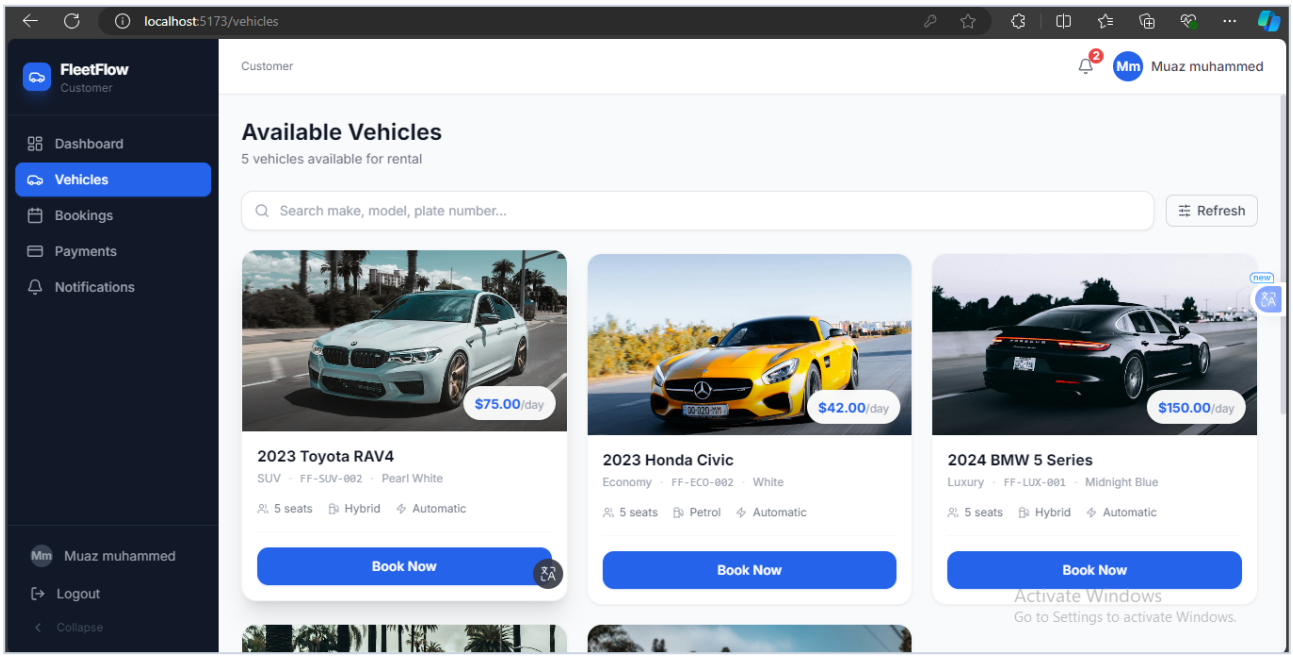
Fig 3.1 — Customer Dashboard (Muaz's Account)



The customer dashboard shows a personalised greeting with the customer's first name and the current date. Three KPI cards display Upcoming Bookings, Active Rentals, and Completed rentals. A "Book a Vehicle" button in the top right provides quick access to the booking form. The main content area shows a list of available vehicles with thumbnail photos, category, registration number, and daily rate. A Quick Actions panel on the right provides direct links to Browse Vehicles, New Booking, My Bookings, and Payment History. The dark sidebar shows only the modules relevant to a customer: Dashboard, Vehicles, Bookings, Payments, and Notifications.

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Fig 3.2 — Available Vehicles: Customer View



The vehicles page shows all currently available vehicles in a responsive card grid. Each vehicle card displays a full-width photograph, the vehicle name, category, registration number, colour, seating capacity, fuel type, transmission type, and the daily rental price in a pill badge on the image. A prominent "Book Now" button appears at the bottom of each card. Customers can search by make, model, or plate number using the search bar at the top. The page shows 5 vehicles available for rental: Toyota RAV4, Honda Civic, BMW 5 Series, Ford Explorer, and Toyota Corolla.

Customer Bookings: BK0STUT4QN

The bookings page lists all of the customer's reservation history in a table. Each row shows the booking reference (a clickable blue link), the vehicle name, pickup date, return date, total amount, and current status badge. A "+ New Booking" button allows customers to create new reservations, and a status filter dropdown filters by booking state. Booking **BK0STUT4QN** for a BMW 5 Series from Oct 10–11, 2026, costing $365.00, carries an **APPROVED** badge — meaning staff have approved it and the customer can now proceed to pay.

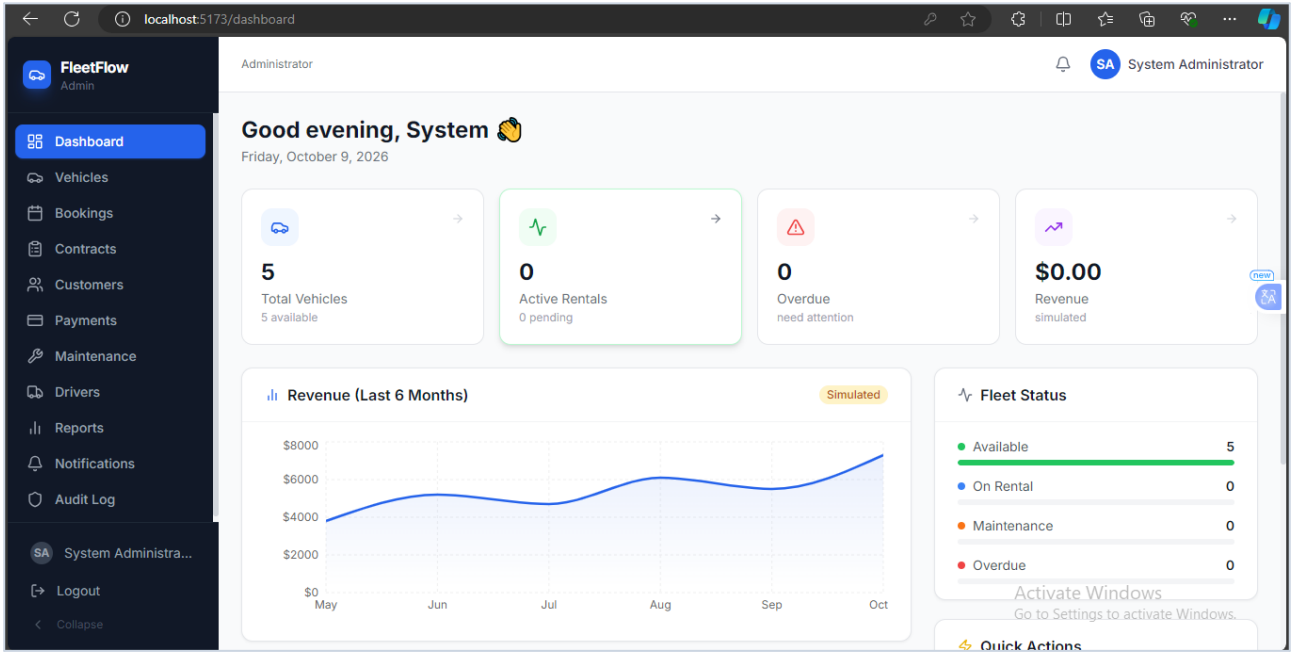
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SECTION 7

Administrator Role — Screens & Features

Full system control with complete visibility across all operations.

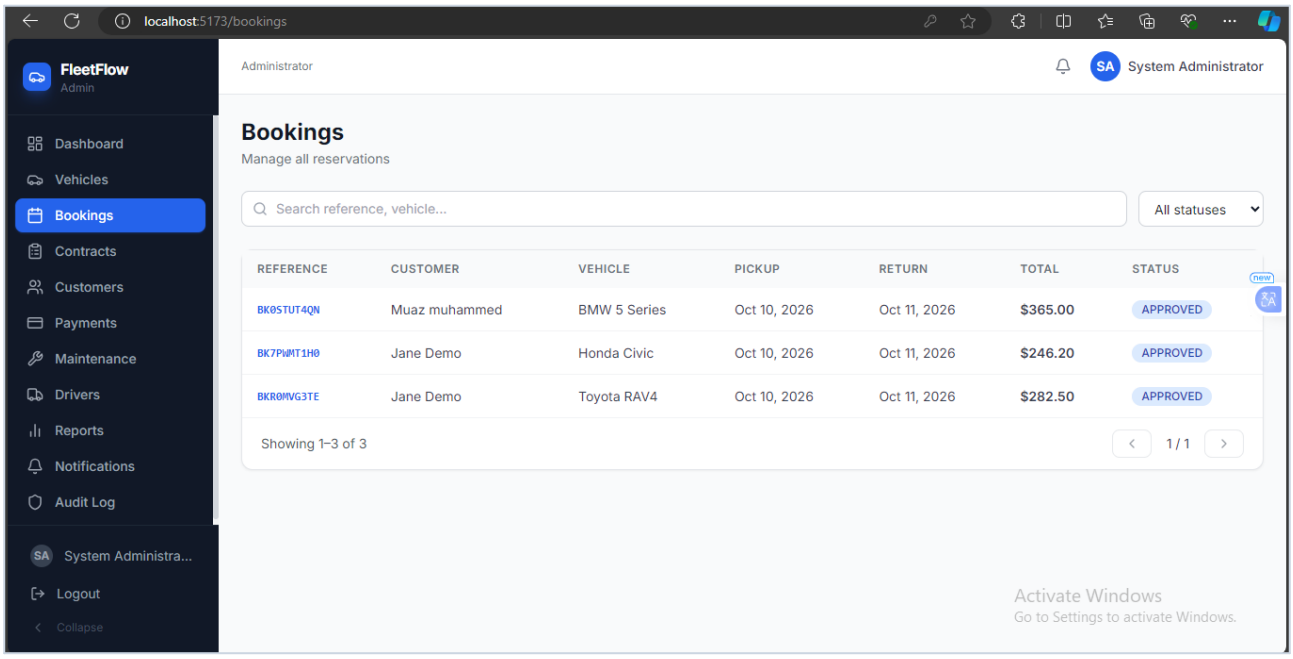
Fig 4.1 — Administrator Dashboard



The dashboard shows a comprehensive overview of the entire system. Four KPI cards display Total Vehicles (5, with 5 available), Active Rentals (0, 0 pending), Overdue (0, need attention), and Revenue ($0.00, simulated). Below is a Revenue area chart spanning the last 6 months with a "Simulated" label. On the right, a Fleet Status panel shows colour-coded progress bars for Available, On Rental, Maintenance, and Overdue counts. The sidebar shows all modules, including Contracts, Customers, Maintenance, Drivers, Reports, and Audit Log — which are not available to customers.

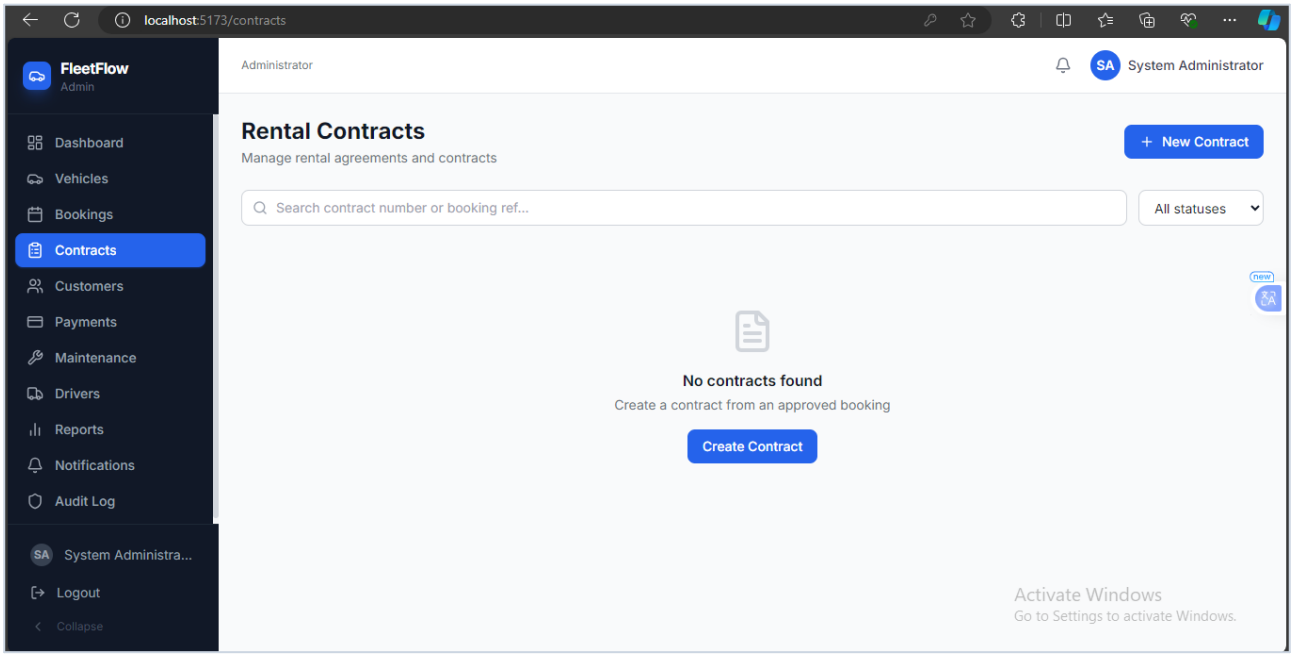
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Fig 4.2 — Admin Bookings Management



Unlike the customer view, the administrator bookings table includes a Customer column. Three bookings are visible: Muaz muhammed booked a BMW 5 Series ($365.00), Jane Demo booked a Honda Civic ($246.20), and Jane Demo also booked a Toyota RAV4 ($282.50). All three are in APPROVED status. Administrators can click any booking to view full details, approve or reject pending bookings, record pickups and returns, and manage the entire booking lifecycle.

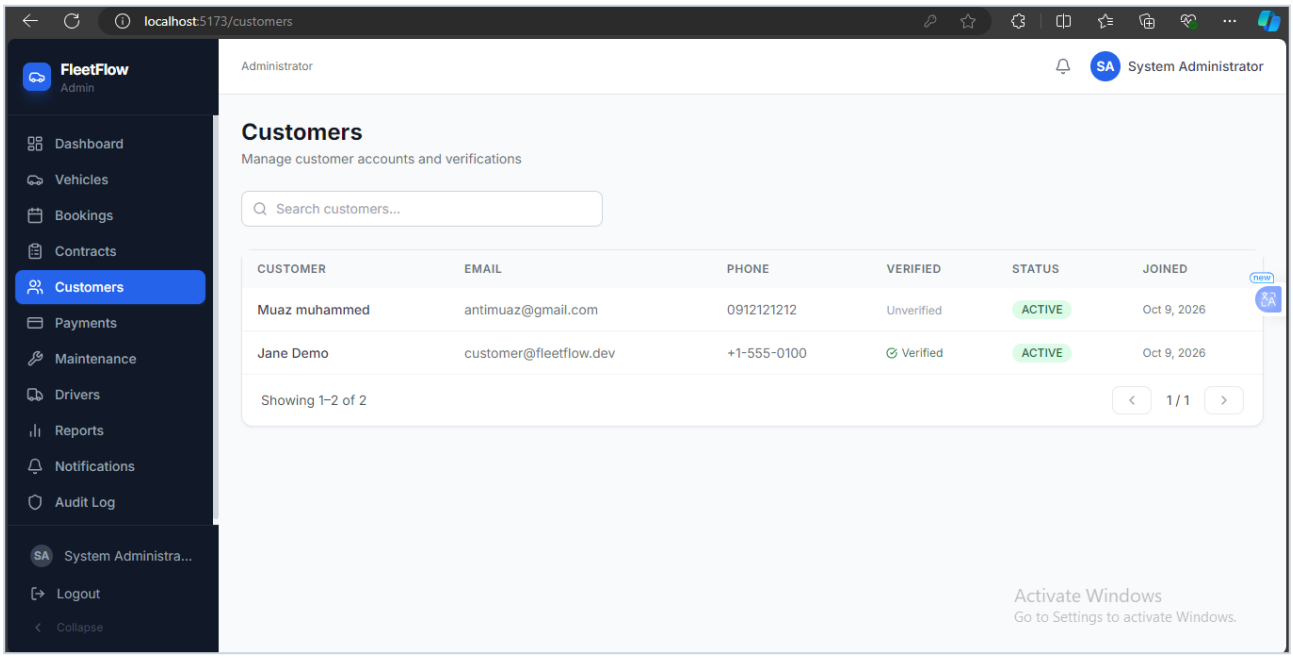
Fig 4.3 — Rental Contracts Page



This page allows staff to generate and manage formal rental agreements. It shows a search bar and status filter, and currently displays "No contracts found — Create a contract from an approved booking". A "+ New Contract" button and a "Create Contract" button are available. To create a contract, staff enter the booking reference of an approved booking, and the system generates a numbered contract (e.g., CT-2026-00001) with standard rental terms, linking it to the booking, customer, and vehicle.

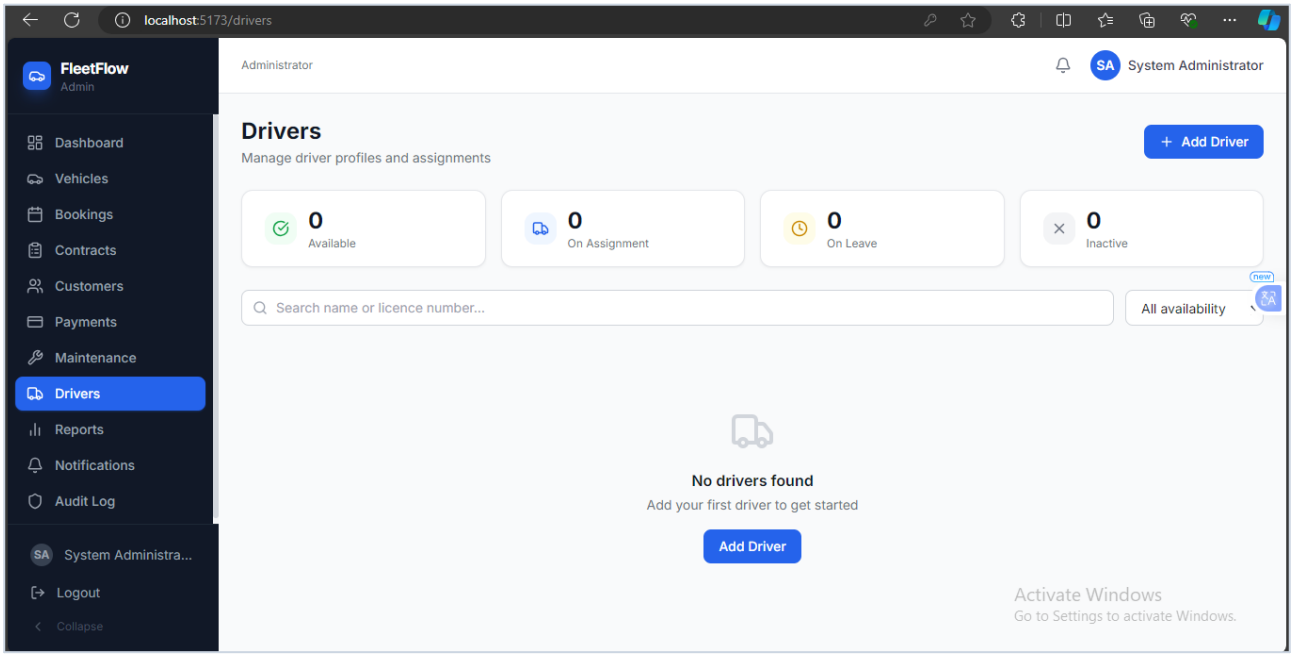
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Fig 4.4 — Customers Management Page



This page lists all registered customers with their name, email address, phone number, verification status, account status, and join date. Two customers are shown: Muaz muhammed (antimuaz@gmail.com, 0912121212, Unverified, Active) and Jane Demo (customer@fleetflow.dev, +1-555-0100, Verified, Active). Staff can click any customer to view their full profile, booking history, and driving licence information, and can mark customers as verified once their identity and licence have been confirmed.

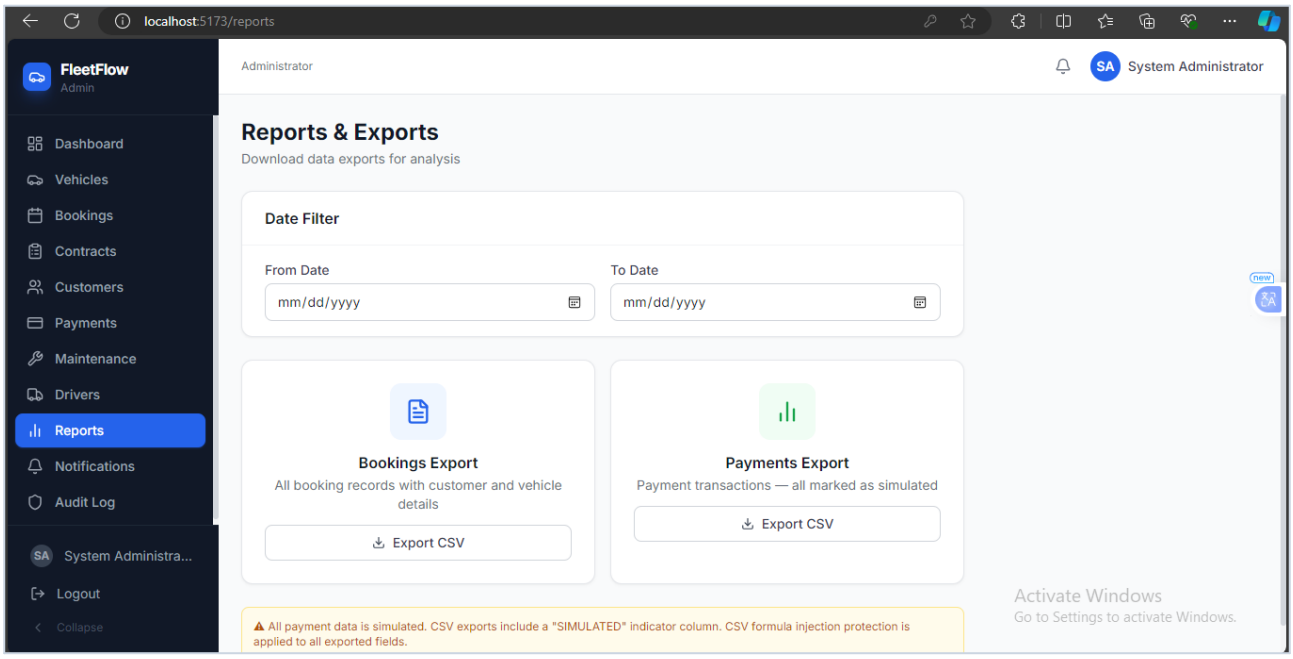
Fig 4.5 — Drivers Management Page



This page manages all registered drivers in the fleet. Four status cards show counts for Available (0), On Assignment (0), On Leave (0), and Inactive (0). No drivers have been added yet, so the empty state "No drivers found — Add your first driver to get started" is displayed with an "Add Driver" button. Clicking it opens a form to enter the driver's name, email, phone, licence number, licence expiry date, licence class, and years of experience.

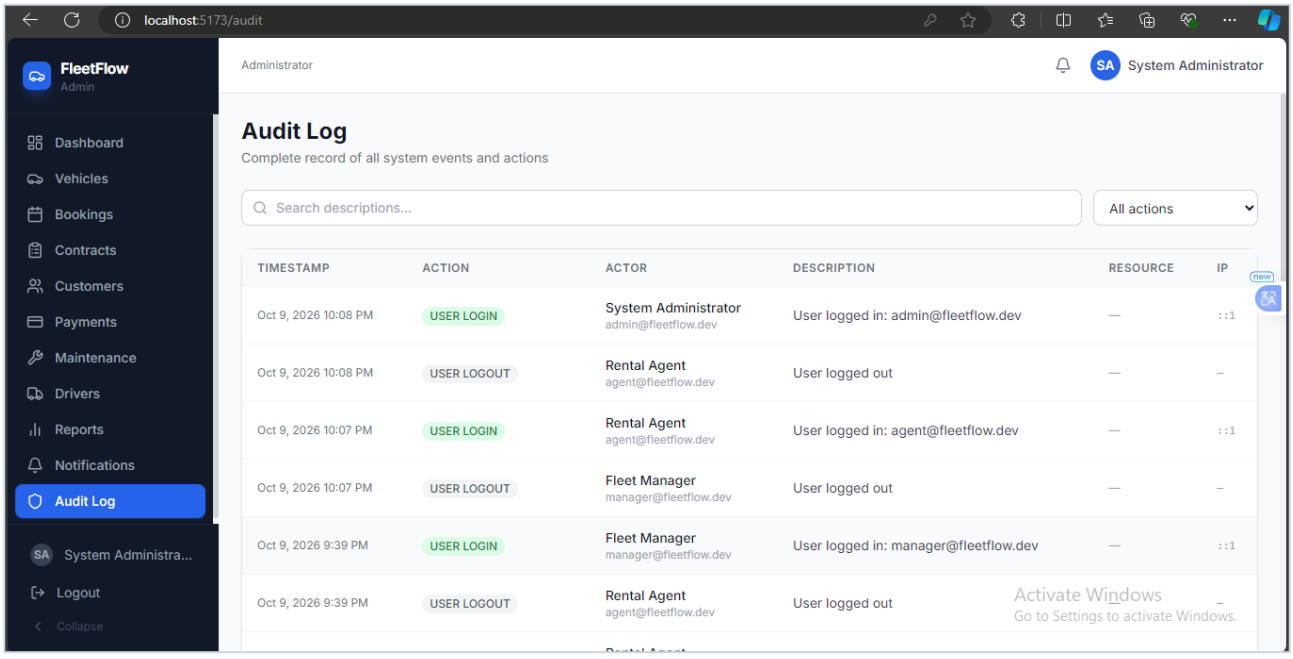
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Fig 4.6 — Reports & Exports Page



This page provides data export for administrators and fleet managers. A date filter with "From Date" and "To Date" fields allows filtering by time period. Two export options are presented: **Bookings Export** (all booking records with customer and vehicle details) and **Payments Export** (payment transactions, all marked as simulated). Both download as CSV files. A warning banner notes that all payment data is simulated and that CSV formula injection protection is applied to all exported fields.

Fig 4.7 — Audit Log (Administrator Only)



The audit log is the system's complete security record, accessible only to administrators. Every significant action is captured with a timestamp, action type, actor name and email, description, affected resource, and IP address. The log shows login and logout events for the System Administrator, Rental Agent, and Fleet Manager accounts. Action types are colour-coded: USER LOGIN (green), USER LOGOUT (grey), BOOKING APPROVED (green), and so on. Staff can filter by action type and search by description.

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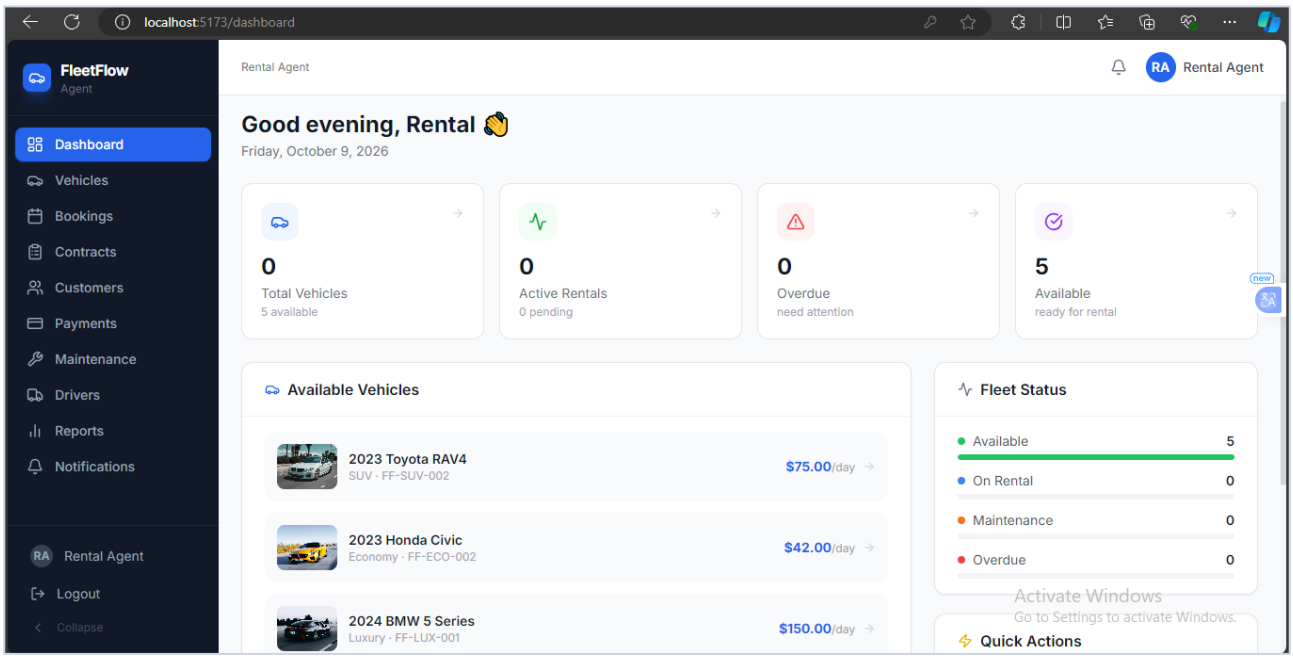
SECTION 8

Rental Agent Role — Screens & Features

The front-desk role focused on daily customer operations.

The Rental Agent is responsible for the operational side of bookings — approving or rejecting customer requests, managing vehicle pickups and returns, recording inspections, and assisting customers with their rentals. Their dashboard is focused on operational data rather than financial analytics.

Fig 5.1 — Rental Agent Dashboard



The agent dashboard shows four KPI cards and an "Available Vehicles" list with thumbnail photos, helping the agent quickly see what is ready for rental. A Fleet Status panel shows colour-coded bars for Available, On Rental, Maintenance, and Overdue, and a Quick Actions panel includes links to View Bookings, Fleet Overview, and Maintenance. Notice that the sidebar does not include Audit Log — it is restricted to administrators.

Agent Responsibilities

A Rental Agent's typical daily workflow in FleetFlow:

1	**Review Pending Bookings** Open the Bookings page and filter by PENDING status. Review each request for valid dates and vehicle availability.
2	**Approve or Reject** Click the booking to open the detail page. Click Approve (notifies customer, who can then pay) or Reject (with an optional reason shown to the customer).
3	**Record Pickup Inspection** When the customer arrives, record the pre-rental inspection — current mileage, fuel level, any existing damage — and confirm pickup. Booking moves to ACTIVE.

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4	**Record Return Inspection** When the vehicle is returned, record the post-rental inspection. The system calculates late-return fees automatically. Booking moves to COMPLETED.
5	**Verify Customers** Check driving licence details on the customer profile and mark the customer as verified once their identity is confirmed.

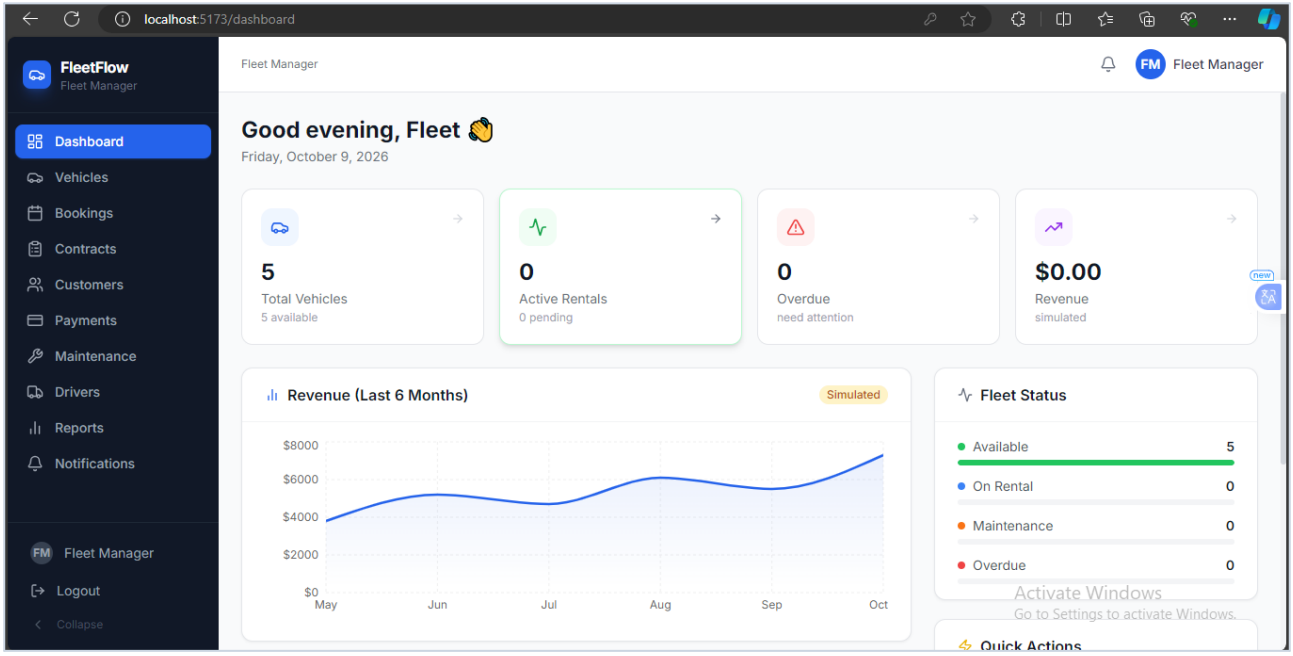
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SECTION 9

Fleet Manager Role — Screens & Features

Strategic fleet oversight, vehicle management, and operational reporting.

Fig 6.1 — Fleet Manager Dashboard



The dashboard is identical in structure to the Administrator dashboard, showing the full set of KPI cards (Total Vehicles, Active Rentals, Overdue, Revenue), the Revenue area chart, and Fleet Status bars. The key difference is that the sidebar does not include the Audit Log — Fleet Managers can manage operations and finances but do not have access to the security event log. The account is labelled "Fleet Manager" in both the sidebar and the top-right header, confirming the role context.

Fleet Manager Responsibilities

The Fleet Manager focuses on the operational and financial health of the fleet:

Vehicle Management
Add new vehicles to the fleet with full specs, pricing, insurance and registration dates. Edit existing vehicles and archive decommissioned ones.

Maintenance Scheduling
Create and schedule maintenance records for vehicles. Assign technicians, track costs, and the system automatically marks vehicles unavailable during active maintenance.

Driver Management
Add driver profiles with licence details, manage availability, assign drivers to bookings, and track assignment history.

Reports & Analytics
View revenue charts, fleet utilisation, and export booking and payment data as CSV files for external analysis.

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Contracts
Generate rental contracts from approved bookings. View contract terms, signing dates, and manage contract status.

Customer Oversight
View all customer profiles, verify driving licences, and manage account status.

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SECTION 10

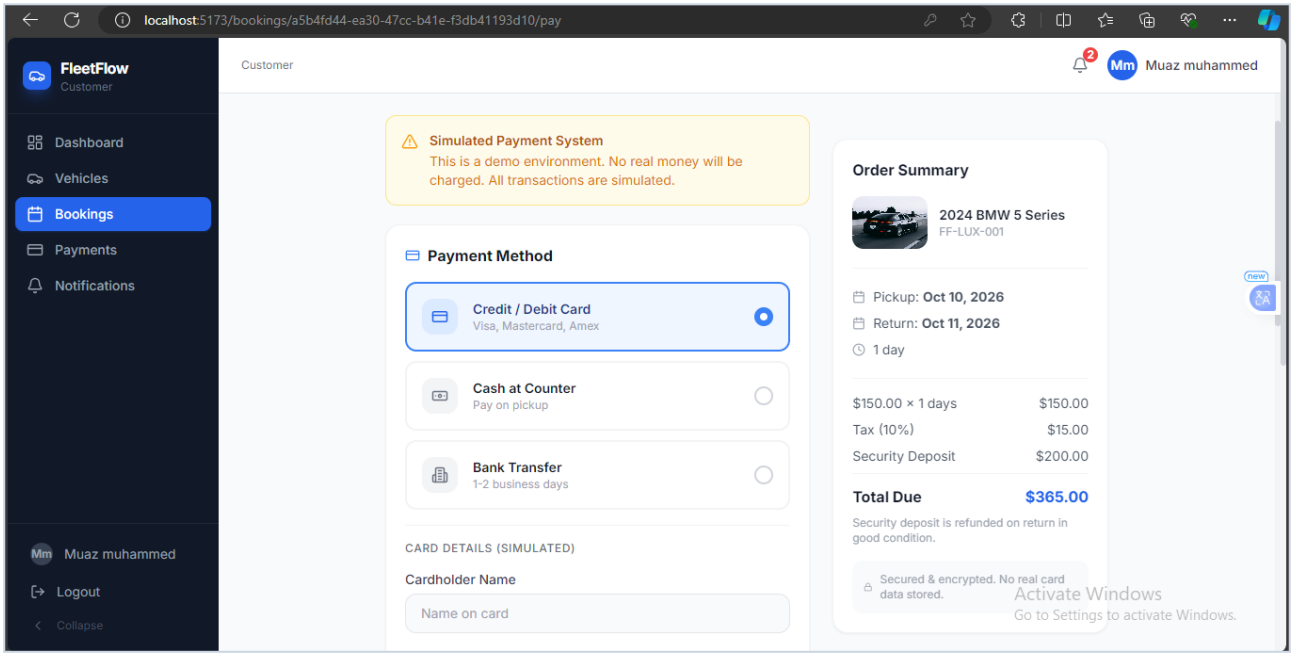
Booking & Payment Lifecycle

The complete end-to-end journey from booking request to rental completion.

FleetFlow models the full rental workflow as a state machine. A booking passes through defined statuses, with each transition triggering notifications, payment requirements, and vehicle status updates.

1	**Customer Submits Booking → PENDING** Customer selects vehicle, dates, and location. The server calculates all pricing (daily rate × days + tax + security deposit) and checks availability using a serializable database transaction to prevent double-booking. Customer receives a notification confirming submission.
2	**Staff Approves Booking → APPROVED** An agent or administrator reviews the booking and approves it. The customer receives a notification and a blue "Payment Required" banner appears on their booking detail page.
3	**Customer Pays → CONFIRMED** Customer opens the payment page, selects a payment method (Card, Cash, or Bank Transfer), provides a security deposit guarantee (Credit Card Hold, ID Copy, or Cash Deposit), agrees to terms, and submits. All payments are clearly marked as SIMULATED. The booking advances to CONFIRMED.

Fig 7.1 — Payment Page: Order Summary & Payment Method



After a booking is approved, the customer reaches this page. A yellow banner states that this is a simulated payment system and no real money is charged. The customer chooses a payment method — Credit / Debit Card, Cash at Counter, or Bank Transfer — and fills in simulated card details. The Order Summary on the right shows the 2024 BMW 5 Series (FF-LUX-001), pickup Oct 10, 2026 and return Oct 11, 2026 (1 day), then the server-calculated breakdown: $150.00 × 1 day = $150.00, Tax (10%) $15.00, Security Deposit $200.00, for a **Total Due of $365.00**. The security deposit is refunded on return in good condition.

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1. **Staff Records Pickup → ACTIVE**
   On the pickup date, a rental agent records the pre-rental inspection (mileage, fuel level, existing damage). The vehicle status changes to RENTED and the booking becomes ACTIVE.

2. **Staff Records Return → COMPLETED**
   When the vehicle is returned, the agent records the post-rental inspection. The system automatically calculates any late return fees (grace period of 2 hours, then 1.5× daily rate per late day). The vehicle status returns to AVAILABLE and the booking is COMPLETED.

3. **Security Deposit Released**
   Once the vehicle is confirmed returned in good condition, the security deposit is released back to the customer. Any damage costs or late fees are charged from the deposit before release.

Pricing transparency: All financial totals are calculated on the server. Customers cannot manipulate prices by modifying API requests — any price submitted by the client is ignored and recalculated server-side.

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SECTION 11

Security Features

Security is built into every layer of FleetFlow, not added as an afterthought.

Security Control	Implementation
Password Storage	Argon2id with 64 MB memory cost, 3 time iterations, 4 parallel threads. Industry's strongest algorithm.
Authentication	JWT access tokens (15-minute expiry) + refresh tokens (7-day expiry). Refresh tokens are stored in the database and can be revoked on logout or password change.
Authorization	Role-based access control enforced on every API endpoint. Frontend UI changes reflect permissions but the backend independently verifies every request.
IDOR Protection	Customers can only access their own bookings, payments, and notifications. Any attempt to access another user's data returns 403 Forbidden.
Rate Limiting	Auth endpoints: 10 attempts per 15 minutes. Password reset: 5 per hour. General API: 100 per 15 minutes.
SQL Injection	All database access through Prisma ORM with parameterized queries. Raw SQL is never used.
Security Headers	Helmet.js sets: Content-Security-Policy, X-Frame-Options, X-Content-Type-Options, Referrer-Policy, and more.
CORS	Restricted to the configured frontend origin only. No wildcard CORS.
Input Validation	All request bodies validated with Zod schemas before processing. Invalid inputs return structured error messages.
CSV Injection	All CSV export fields are sanitised. Values starting with =, +, -, @, tab, or carriage return are escaped to prevent formula injection in spreadsheet applications.
Concurrency Safety	Booking creation uses PostgreSQL serializable isolation level to prevent double-booking under concurrent requests.
Audit Logging	All logins, logouts, password changes, booking approvals, payments, and admin actions are logged with timestamp, actor, IP address, and resource reference.
Secret Management	All secrets (JWT keys, database credentials) are in .env files which are gitignored. No secrets are committed to the repository.
Production Safety	Stack traces are suppressed in production responses. Error messages are generic to prevent information leakage.
Payment Security	Only the last 4 digits of a card are ever stored as a display reference. No card numbers, CVVs, or full payment credentials are stored anywhere.

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SECTION 12

Conclusion

FleetFlow is a complete, production-quality vehicle rental and fleet management platform that demonstrates real-world software engineering across every discipline — database design, REST API architecture, secure authentication, business logic, role-based access control, and modern frontend development.

What Was Delivered

- A responsive public landing page with animations, vehicle showcase, and clear CTAs
- A secure authentication system with Argon2id hashing, JWT tokens, rate limiting, and password reset
- A complete vehicle fleet module with category management, compliance tracking, and image support
- A concurrency-safe booking system that prevents double-booking using serializable transactions
- A realistic payment flow with security deposits, collateral options, and simulated transactions
- A full rental lifecycle: submit → approve → pay → pickup inspection → active → return inspection → complete
- A maintenance management module with scheduling, technician assignment, and vehicle status integration
- A driver management module with licence tracking and assignment conflict detection
- A contracts module that generates formal rental agreements from approved bookings
- A reports module with CSV export and injection protection
- A complete audit log recording all system events for security and compliance
- A role-specific dashboard for each of the four user roles
- A real PostgreSQL database with 20+ entities, foreign keys, indexes, and migrations
- 13 automated tests covering the pricing engine and authentication utilities
- Docker and GitHub Actions configuration for containerised deployment and CI/CD

Known Limitations & Future Work

- Payments are simulated — a real payment provider (e.g. Stripe) would need to be integrated for production
- Email uses Ethereal preview in development — production requires a real SMTP provider
- No HTTPS enforcement — TLS should be configured at the reverse proxy level before deployment
- Rate limiting uses an in-memory store — Redis should be used for multi-instance deployments
- File uploads for licence documents and vehicle images are scaffolded but require cloud storage (e.g., S3) for production

> **Access the system:** Frontend → http://localhost:5173 | Backend API → http://localhost:4000/api | Health Check → http://localhost:4000/health

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